The Causal Effect of the Americans with Disabilities Act (1990) on Healthcare Access

Tanisha Gauns

April 2026

Abstract

The Americans with Disabilities Act (1990) was landmark legislation that prohibited discrimination and required physical access accommodations across public and private institutions, including healthcare settings. But whether these mandates actually translated into more healthcare use for severely disabled people remains an open empirical question. This paper tries to answer that: specifically, I estimate the causal effect of the ADA on physician utilization among severely disabled working-age adults in the United States.

Using the National Health Interview Survey Activity Limitation Supplement (1983-1996, $N = 88{,}814$), I employ a difference-in-differences design that compares individuals with severe disabilities, those unable to perform any major activity ($\texttt{LATOTAL} = 1$), to those with more moderate limitations ($\texttt{LATOTAL} = 2$), before and after the ADA's passage in 1990. Seven years of pre-policy data allow me to formally test the parallel trends assumption.

My findings suggest the ADA had no detectable effect on whether a severely disabled individual saw any physician in a given year ($\beta = -0.0001$, $p = 0.99$). It did, however, significantly increase the number of physician visits per year by roughly 0.77 ($p = 0.005$), an effect that grows as individual-level controls are added. Pre-treatment parallel trends hold across a joint F-test ($F = 0.28$, $p = 0.95$), which supports the validity of the design.

Taken together, these results suggest the ADA improved the depth of care for severely disabled individuals, enabling more visits among those already engaged with the healthcare system, but did not meaningfully increase initial healthcare entry. Physical access mandates alone appear insufficient to close the extensive margin gap in physician utilization.

Introduction

In the years before the Americans with Disabilities Act, working-age adults with severe physical disabilities faced a paradox that was both well-documented and largely unaddressed. Despite carrying substantially higher burdens of chronic illness, conditions that generate precisely the kind of recurrent, ongoing need for medical attention that the healthcare system is designed to serve, severely disabled Americans reported going without physician care at rates that reflected not medical choice but physical exclusion (LaPlante, 1988). Inaccessible exam tables, office layouts that could not accommodate wheelchairs, and the absence of communication aids routinely turned a routine medical appointment into an insurmountable obstacle. The group with the greatest clinical need was the least likely to cross the threshold of a physician's office. Congress, in 1990, moved to change that.

The Americans with Disabilities Act was designed, in part, as a healthcare access law. Title II took effect immediately, prohibiting discrimination by public entities including public hospitals and health clinics. Title III, which took effect January 26, 1992 after an 18-month implementation period, extended those obligations to private medical providers, mandating physical accessibility modifications, auxiliary aids such as sign-language interpreters, and the elimination of policies that had routinely excluded or segregated disabled patients (U.S. Department of Justice, 1992). The architects of the ADA expected these requirements to work on two margins simultaneously: drawing in disabled individuals who had never been able to access a physician at all, and enabling those already receiving some care to visit more frequently and more completely. Remove the barrier, the logic went, and utilization would follow.

Whether utilization actually followed is what this paper tries to figure out. Studies documenting persistent gaps in physician contact between disabled and non-disabled Americans in the post-ADA period, including Krahn et al. (2006) and Iezzoni et al. (2000), cannot tell us whether those gaps were narrowed, widened, or left unchanged by the law. Without baseline data from before 1990, a disparity observed in 1995 is consistent with three entirely different stories: the ADA failed; the ADA helped but the pre-existing gap was even larger; or the ADA had nothing to do with it and secular trends explain whatever movement occurred. Cross-sectional comparisons also conflate the law's effect with every other way in which disabled and non-disabled Americans differ. Separating the ADA's contribution from that background requires data that straddles the policy change and none of the prior healthcare studies have it.

This paper fills that gap. Using the National Health Interview Survey Activity Limitation Supplement from 1983 to 1996, I apply a difference-in-differences design that compares severely disabled working-age adults to a comparison group of adults with moderate limitations, before and after the ADA's enactment. These two groups are identified using the survey variable `LATOTAL`, which classifies respondents by their degree of activity limitation: a value of 1 (`LATOTAL` = 1) denotes complete inability to perform any major activity, while a value of 2 (`LATOTAL` = 2) denotes being limited in the kind or amount of major activity one can perform. The analytic sample covers 88,814 individuals observed across 14 years. The design's critical advantage over prior work on the ADA and healthcare is seven years of pre-policy data, which makes it possible to formally test whether the two groups were on parallel trends before the law changed, rather than simply assuming it. I estimate effects separately for whether an individual saw any physician in a given year, the extensive margin and how many visits they made, the intensive margin. Figure 1 documents the full sample construction, including how the 88,814-observation analysis sample is derived from the broader NHIS.

The ADA did not significantly increase the probability that a severely disabled individual saw any physician ($\beta = -0.0001$, $p = 0.99$). It did significantly increase the number of physician visits per year by roughly 0.77 ($p = 0.005$), an estimate that grows in magnitude and stability as individual-level controls are added. Parallel trends hold cleanly across all seven pre-ADA years ($F = 0.28$, $p = 0.95$). These findings survive age-group stratification, alternative approaches to missing data, and an alternative treatment cutoff using Title III's 1992 effective date. The headline result, a precise null at the extensive margin alongside a positive intensive-margin effect, is consistent with a law that deepened care for severely disabled individuals already engaged with the healthcare system, while doing relatively little to bring those outside the system in.

Literature Review

The most rigorous causal work on the ADA addresses employment rather than health. Acemoglu and Angrist (2001) apply a difference-in-differences design to Current Population Survey data and find that the ADA actually reduced employment among disabled men, a result they attribute to employers factoring in the higher cost of accommodation mandates when making hiring decisions. DeLeire (2000) arrives at a similar conclusion, documenting a post-ADA employment decline among disabled workers using comparable data. Bound and Waidmann (2002) are more skeptical, arguing that this decline predates the ADA and reflects broader labor market shifts rather than anything specific to the law , a useful reminder that pre-existing trends can easily be mistaken for policy effects if researchers are not careful.

These labor market findings matter for the present study in two ways. First, they show that the ADA did change real behavior: if employers responded to the law by adjusting how they treated disabled workers, there is reason to think healthcare providers may have done the same with disabled patients. Second, they set a sobering precedent. The ADA's employment effects were, at best, negligible and, at worst, counterproductive. This should temper any assumption that a sweeping anti-discrimination law will automatically deliver the outcomes it was designed to produce.

A separate body of work documents persistent healthcare access gaps between disabled and non-disabled Americans, though nearly all of it relies on post-ADA cross-sectional data. Iezzoni et al. (2000) find that adults with mobility impairments are significantly less likely to receive routine preventive care, cancer screenings, vaccinations, and the like, than their non-disabled peers, even after controlling for insurance and sociodemographic factors. Krahn et al. (2006) document a cascading pattern of disparities for people with intellectual disabilities, showing that limited access compounds across multiple dimensions of care, from preventive services to specialist referrals. Krahn et al. (2015) push this further, arguing that people with disabilities represent an overlooked health disparity population whose access gaps are systematically undercounted in national data, partly because disability status is rarely treated as a demographic category alongside race or income.

More recent work has begun to link these access gaps directly to the ADA. Parker Harris et al.(2017) conducted a systematic review of the evidence on the ADA and healthcare access and found that, despite Title III's broad anti-discrimination mandate, disabled individuals continue to face structural and attitudinal barriers in clinical settings that the law has not resolved. Parker Harris et al. (2019) summarize this evidence in a research brief, noting that physical inaccessibility of facilities, provider communication gaps, and insurance-related exclusions persist more than two decades after the law's passage. Ordway et al. (2021) offer a mixed-methods perspective, documenting that both patients and providers perceive significant ADA compliance gaps in healthcare settings, with providers often unaware of their obligations under Title III.

These studies make clear that a gap exists. What they cannot tell us is whether the ADA had anything to do with it. Because all of these studies draw on post-ADA data alone, a disparity observed in any year after 1990 could reflect three very different realities: the ADA failed to close a gap that was already there, the ADA helped but the original gap was simply too large to eliminate, or the ADA had no effect at all and the disparity is driven by something else entirely. Comparing disabled and non-disabled individuals only makes this harder to untangle, since the two groups differ in so many ways beyond disability status that isolating the law's specific effect becomes nearly impossible.

Two gaps in the prior literature motivate this paper:

1. **Absence of pre-ADA baseline data.** Without pre-ADA trends, the parallel trends assumption underlying a causal interpretation cannot be tested, it can only be asserted. LaPlante (1988), drawing on the National Health Interview Survey's Activity Limitation Supplement, provides one of the few systematic accounts of healthcare utilization among disabled Americans before the ADA's passage. The seven years of pre-ADA NHIS Activity Limitation Supplement data used here build on that foundation, making it possible to verify, rather than assume, that severely and moderately disabled individuals were on comparable healthcare trajectories before 1990.

2. **Treatment of utilization as a single outcome.** Prior studies measure whether a person had any physician contact or reported any unmet need,

collapsing two economically distinct responses: whether the ADA drew people into the healthcare system who had previously been excluded (the extensive margin), and whether it enabled those already receiving care to access it more frequently or more completely (the intensive margin). These margins have different policy implications, the first speaks to access barriers, the second to care continuity, and the existing literature does not separate them.

This paper addresses both gaps. It is the first to use pre-1990 NHIS data to formally test parallel trends in healthcare utilization between disability severity groups before the ADA, and the first to decompose the ADA's effect into extensive and intensive margin responses. The within-disability severity design, comparing severely disabled individuals to those with moderate limitations rather than disabled to non-disabled, further avoids the confounds that arise when the comparison group differs from the treatment group on many dimensions beyond disability status.

Data

The primary dataset is the National Health Interview Survey (NHIS), accessed through IPUMS NHIS (Blewett et al., 2024). The NHIS is an annual cross-sectional household survey conducted by the National Center for Health Statistics, designed to track the health of the U.S. civilian non-institutionalized population. The study covers 1983 to 1996, seven pre-ADA years (1983–1989) and six post-ADA years (1990–1996). The start date is set by the first year the Activity Limitation Supplement was fielded; the end date reflects a data comparability constraint, as the NHIS underwent a comprehensive redesign in 1997 that replaced the supplement with a new disability classification instrument, making post-1996 disability codes fundamentally incompatible with the pre-1997 baseline. The full cleaned NHIS extract spans 1983–2000 and contains 1,130,227 working-age (18–64) person-year observations; the study window of 1983–1996 accounts for 889,505 of these.Of the 1983–1996 observations, 88,814,roughly 10.0 percent have a valid Activity Limitation Supplement response (LATOTAL code 1 or 2) and make up the analysis sample. The remaining respondents were not administered the supplement in their survey year, a deliberate design feature rather than a data gap. Respondents coded as not-in-universe for the supplement (`LATOTAL` = 0) are excluded entirely, since their disability status is unobserved and treating their absence as evidence of no disability would introduce systematic measurement error into the treatment variable. Supplement selection rates hold steady between 9.4 and 10.9 percent across years with no noticeable shift around 1990, and the share of supplement respondents classified as severely disabled rises gradually from 40.3 percent in 1983 to 53.8 percent in 1996 with no visible break at the ADA's enactment, ruling out strategic reclassification into the treatment group after the law took effect (Figure 2). Table 1 documents the full sample construction.

The analysis relies on three core variables. The disability severity variable `LATOTAL`, drawn from the Activity Limitation Supplement, classifies respondents as severely disabled (`LATOTAL` = 1; unable to perform their major activity; N = 42,294) or moderately limited (`LATOTAL` = 2; limited in kind or amount of activity; N = 46,520), forming the treatment and control groups respectively. This is a within-disability severity design, the control group is not the non-disabled population. The logic is that the ADA's physical access mandates bind most tightly for individuals who cannot perform major activities at all; those with moderate limitations were already better positioned to access care without formal structural accommodation, making severity the natural source of identifying variation. Two outcomes capture different dimensions of healthcare utilization. The first, `saw_doctor`, is a binary indicator equal to 1 if the respondent visited any physician at least once in the past 12 months, constructed from NHIS variable `DV12`; available for 99.7 percent of the analysis sample (N = 88,505), it measures the extensive margin whether the ADA changed whether severely disabled individuals entered the healthcare system at all. The second, `dv12_clean`, counts physician visits in the past 12 months capped at 96, with administrative placeholder codes (97 and above) recoded to missing; available for 99.6 percent of the sample (N = 88,485), it measures the intensive margin, how often severely disabled individuals saw a physician among those already using care. Table 1- Panel B documents outcome coverage by year.

The NHIS is used for the main causal analysis for two reasons. First, it is the only nationally representative survey that administered a consistent disability severity question across both pre- and post-ADA years, providing seven pre-ADA baseline years needed to empirically test whether the treatment and control groups were on parallel trajectories before the law changed, something no prior ADA healthcare study has had access to. Second, its large annual samples, with supplement selection rates holding steady between 9.4 and 10.9 percent across years with no noticeable shift around 1990, support regression-based difference-in-differences estimation at a scale the Current Population Survey and Behavioral Risk Factor Surveillance System simply cannot match. The BRFSS plays a secondary role in the analysis: it serves as an external validity check on the disability-healthcare gap found in the NHIS, provides descriptive state-level evidence on healthcare access by disability status, and supports sensitivity checks on routine checkup rates unavailable in the post-supplement NHIS. Because the BRFSS begins only in 1994 four years after the ADA's enactment, it has no meaningful pre-ADA baseline and cannot anchor any causal estimates. All causal results in this paper come from the NHIS.

Methodology

Difference-in-Differences Design

The central empirical challenge in estimating the ADA's effect on healthcare access is that the severely disabled and moderately limited differ in many ways beyond how much the law affected them. A simple before-and-after comparison, how did the severely disabled change after 1990? would confound the ADA's effect with any other trend affecting healthcare utilization over the same period. A simple cross-sectional comparison, do the severely disabled use more care than the moderately limited? would pick up pre-existing differences between the groups rather than anything the ADA actually caused. The difference-in-differences (DiD) design addresses both problems at once by combining the two comparisons: it asks whether the severely disabled changed more than the moderately limited after the ADA, relative to how the two groups were already diverging or converging before it. That double difference the change in the gap is what gets attributed to the law.

The estimating equation is:

$$Y_{it} = \alpha + \beta\,(\text{Treated}_i \times \text{Post}_{1990_t}) + \gamma\,\text{Treated}_i + \delta\,\text{Post}_{1990_t} + \lambda_t + X_i'\phi + \varepsilon_{it} \quad (1)$$

where Y_{it} is the healthcare outcome for person i in year t; $\text{Treated}_i = 1$ if `LATOTAL` $= 1$ (unable to perform major activity) and 0 if `LATOTAL` $= 2$ (limited in kind or amount); $\text{Post}_{1990_t} = 1$ if the survey year is 1990 or later; λ_t are year fixed effects that absorb aggregate time trends affecting both groups equally; and X_i is a vector of individual-level controls. The coefficient β is the parameter of interest, the average treatment effect of the ADA on the severely disabled relative to the moderately limited.

Three specifications are estimated for each outcome, using weighted least squares (WLS) with NHIS person weights and HC3 heteroskedasticity-robust standard errors throughout:

- **Model 1 — Raw DiD:** no year fixed effects, no controls. This is the most transparent specification and sets a baseline against which the other models can be compared.
- **Model 2 — DiD with year fixed effects:** absorbs aggregate trends that affect both groups equally, such as the general expansion of managed care and Medicaid during the early 1990s.
- **Model 3 — DiD with year fixed effects and individual controls:** the preferred specification. Controls include age, sex, self-reported health status, census region, and marital status. Education and employment status are excluded because both are missing for more than 97 percent of the Activity Limitation Supplement subsample, making their inclusion infeasible. Health insurance coverage (`any_insurance`) is missing for 46 percent of the sample and is excluded from the baseline, but tested as a robustness

check using the subsample for which it is observed. The remaining controls age, sex, health status, region, and marital status are complete for 97–100 percent of the sample, with the exception of above-poverty (12.5% missing); listwise deletion on this variable reduces the Model 3 sample from 88,505 to 78,008, a loss of roughly 11 percent. Robustness checks confirm the estimates are insensitive to how this missingness is handled.

The analysis sample covers 88,814 person-year observations drawn from the 1983–1996 NHIS Activity Limitation Supplement, of which 42,294 are severely disabled (`LATOTAL` = 1) and 46,520 are moderately limited (`LATOTAL` = 2).

The Parallel Trends Assumption

The DiD estimator identifies the causal effect of the ADA only if, in the absence of the law, the severely and moderately disabled would have followed the same trend in healthcare utilization over time. This is the parallel trends assumption, and it is the identifying assumption on which the entire design rests.

The assumption cannot be directly tested in the post-ADA period, the law was in place, so any post-1990 divergence between the groups could reflect either the ADA's effect or a pre-existing differential trend. What can be tested is whether the two groups were already diverging in the pre-ADA years before the law changed. Figure 4 plots the weighted group means for both groups across the pre-ADA years (1983–1989), giving a visual impression of whether the two series were tracking each other before the law changed. The following regression is estimated on the pre-period subsample (1983–1989) only:

$$\texttt{saw\_doctor}_{it} = \alpha + \sum_{j=1984}^{1989} \beta_j \left(\text{Treated}_i \times \mathbf{1}[\text{year} = j]\right) + \gamma\,\text{Treated}_i + \lambda_t + \varepsilon_{it} \quad (2)$$

This produces six year × Treated interaction coefficients, one for each pre-ADA year relative to the 1983 base year, capturing whether the gap between severely and moderately disabled individuals was widening or narrowing in any given pre-policy year. A joint F-test then asks whether all six interaction coefficients are simultaneously equal to zero.

The result is unambiguous: $F = 0.276$, $p = 0.949$. In plain terms, a p-value of 0.949 means there is a 94.9 percent probability of observing data like these if the two groups were genuinely trending in parallel before 1990. The test finds no evidence whatsoever of pre-existing differential trends. Figure 5 plots each of the six coefficients with its 95 percent confidence interval; all six overlap zero comfortably, and the pattern shows no systematic drift in either direction. Looking at the individual coefficients tells the same story, they range from +0.0005 to +0.0133 in magnitude, all close to zero and none statistically distinguishable from zero (all $p > 0.26$). The severely and moderately disabled were not moving apart before the ADA. This is the strongest empirical support available for the identifying assumption, and it makes the post-1990 DiD estimates credible in a way that prior descriptive work on the ADA and healthcare, which lacks any pre-period baseline cannot claim.

One limitation is worth acknowledging directly: parallel trends testing is only feasible for `has_regular_doc` and `delayed_care`, the parallel trends assumption cannot be empirically assessed because neither variable is observed in the NHIS before 1990. Three theoretical considerations nonetheless support its plausibility. First, the same two groups exhibit strong parallel pre-trends on `saw_doctor` ($F = 0.276$, $p = 0.949$), suggesting the severely and moderately disabled were not diverging in their healthcare behavior before the ADA. Second, no pre-1990 policy specifically targeted $\texttt{LATOTAL} = 1$ individuals on dimensions of care continuity or delayed access, removing the most obvious mechanism for differential pre-trends. Third, the socioeconomic and demographic factors driving care continuity and delayed care , income, insurance coverage, geography affected both groups symmetrically in the pre-ADA period. Given that the assumption cannot be verified, results for these two outcomes are reported as post-ADA descriptive cross-sections rather than causal DiD estimates and should be interpreted accordingly.

Event Study

Figure 6 presents the event study estimates across both panels. As a further visual check on identification, I estimate an event study by replacing the single post indicator with a full set of year × Treated interaction terms:

$$Y_{it} = \alpha + \sum_{j \neq 1983} \beta_j \left(\text{Treated}_i \times \mathbf{1}[\text{year} = j]\right) + \gamma\, \text{Treated}_i + \lambda_t + \varepsilon_{it} \qquad (3)$$

This traces the treatment effect year by year from 1984 to 1996, with 1983 as the omitted base year. The pre-ADA coefficients (1984–1989) should cluster near zero, visually confirming what the F-test already shows in a single number. The post-ADA coefficients (1990–1996) then reveal the time path of any effect — if provider compliance with the ADA's physical access mandates deepened gradually over the early 1990s as accommodation infrastructure was built out, the intensive-margin coefficients for `dv12_clean` should grow in magnitude over the post-period rather than appearing as a single discrete jump at 1990.

Threats to Identification

Reclassification bias. If individuals strategically shifted from $\texttt{LATOTAL} = 2$ to $\texttt{LATOTAL} = 1$ after the ADA in order to qualify for accommodation benefits, the treatment group's composition would change in ways unrelated to the law's healthcare effects, and β would be contaminated by selection rather than reflecting a genuine causal response. Figure 2 rules this out directly: the share of supplement respondents classified as severely disabled rises gradually and continuously from 40.3 percent in 1983 to 53.8 percent in 1996, with no visible break or acceleration at 1990. Strategic reclassification would produce a sharp discontinuity right at the policy date; what the data show instead is a smooth, uninterrupted trend.

Concurrent policies. Medicaid expansions and broader changes in the managed care landscape during the early 1990s could independently increase healthcare utilization and contaminate the post-ADA estimates. The within-disability severity comparison largely differences these out: Medicaid expansions affected both the severely and moderately disabled, not one group selectively. As long as no major early-1990s health policy targeted `LATOTAL` $= 1$ individuals specifically and none did concurrent policies cancel out in the DiD difference.

Supplement sampling changes. If the NHIS altered its supplement sampling protocol around 1990, the composition of the analysis sample could shift independently of the ADA and generate a spurious treatment effect. Figure 3 addresses this directly: the supplement selection rate is stable at 9.4–10.9 percent across all years from 1983 to 1996, with no detectable shift at 1990. The sample is observationally comparable across the pre- and post-policy periods.

Control group contamination. The moderately limited (`LATOTAL` $= 2$) are also covered by the ADA's provisions. If the law improved their healthcare access too, then the control group is itself partially treated, and the DiD estimate understates the true effect for the severely disabled. This is an inherent limitation of any within-group severity design, not a fixable data problem. The estimates in this paper should therefore be read as conservative lower bounds on the full ADA treatment effect rather than precise point estimates.

Internal Validity

Internal validity asks whether β captures the causal effect of the ADA or whether it is contaminated by something else. Three conditions must hold simultaneously: the parallel trends assumption must be satisfied; the treatment and control groups must not be differentially exposed to other concurrent changes; and the composition of the groups must not shift endogenously around the policy date.

This design addresses each condition directly. On parallel trends, the joint F-test on the six pre-period year $\times$ Treated interactions yields $F = 0.276$, $p = 0.949$, among the strongest pre-trend test results possible. Individual pre-period coefficients are all small in magnitude and statistically indistinguishable from zero. The two groups were on parallel healthcare trajectories for the full seven years before the ADA changed anything. On concurrent changes, Medicaid expansions and the broader expansion of managed care in the early 1990s are the most plausible confounders, and both are differenced out by the within-disability comparison because neither policy distinguished between `LATOTAL` $= 1$ and `LATOTAL` $= 2$ individuals. On group composition, Figure 2 provides direct evidence against strategic reclassification: the severity share rises smoothly across the entire 1983–1996 window with no discontinuity at 1990.

One internal validity concern cannot be fully resolved: the moderately limited are themselves covered by the ADA, making the control group partially treated. If the ADA also improved their healthcare utilization, the DiD understates the true effect for the severely disabled. This is a known limitation of any within-group severity design, not a failure of the empirical strategy. The appropriate interpretation is that β is a lower bound on the causal effect of the ADA on the severely disabled and even by that conservative standard, the intensive-margin estimate of approximately 0.77 additional physician visits per year points toward a real, if modest, effect on care frequency among those already in the healthcare system.

Results

Figure 7 presents the 2×2 DiD decomposition for both margins. The severely disabled increased their physician visits per year by more than the moderately limited after the ADA, producing a positive intensive-margin DiD, while the two groups moved in near-lockstep on the extensive margin, consistent with the null result reported below.

Extensive Margin: No Effect on Healthcare Entry

The ADA did not increase the probability that severely disabled individuals saw any physician in a given year, relative to those with moderate limitations. Table 4 presents estimates across three specifications; all are near zero and statistically insignificant. The preferred estimate (Model 3, with year fixed effects and individual controls) is $\hat{\beta} = -0.0001$ (SE = 0.0047), $p = 0.99$. Against a pre-ADA physician visit rate of 90.2 percent among the severely disabled, a change of -0.01 percentage points is economically negligible. The 95% confidence interval, roughly ± 0.92 pp rules out any positive ADA effect larger than approximately one percentage point. With 78,008 observations, the design is well-powered to detect a meaningful effect if one existed. It did not.

The ADA, in short, did not bring new patients into the healthcare system. Accommodation mandates wheelchair ramps, accessible examination tables, sign-language interpreter requirements reduce friction for people already going to appointments. For someone who has never established care with a physician, those physical improvements do not address the earlier barriers of cost, insurance coverage, and finding a willing provider. The law could not solve what it was not designed to solve.

Intensive Margin: Positive Effect on Visit Frequency

The picture changes entirely for visit frequency. Table 5 presents difference-in-differences estimates for physician visits per year, not just whether someone saw a doctor, but how many times they did. Panel A uses `dv12_clean`, the preferred capped measure, which counts the number of physician visits a respondent made in the past 12 months, with values of 97 and above recoded to missing because the NHIS uses those codes as administrative flags rather than genuine visit counts. Panel B uses the uncapped raw measure as an upper bound. The preferred estimate, Model 3 of Panel A, is $\hat{\beta} = +0.77$ (SE = 0.28), $p = 0.005$. The ADA increased annual physician visits among the severely disabled by roughly 0.77 visits relative to the moderately limited a 5.7 percent increase against the pre-ADA mean of 13.5 visits per year. The uncapped upper bound ($\hat{\beta} = +2.87$, SE = 0.97, $p = 0.003$) is substantially larger precisely because including the administrative codes inflates both the variance and the point estimate; it is reported for completeness, not interpretation.

Two features of the intensive margin results strengthen the causal reading. First, the estimate grows in magnitude as controls are added from +0.6203 in the raw specification to +0.77 in the preferred one, the opposite of what confounding would produce. Second, the result is statistically significant at the 5 percent level in the preferred specification and at 1 percent in the uncapped model. The direction, magnitude, and significance are consistent across every column of Table 5.

A null extensive margin alongside a positive, significant intensive margin is the paper's central finding, and the combination is more informative than either result alone.

The ADA deepened care for individuals already in the system. It did not open the door for those outside it. This asymmetry makes sense mechanically: a wheelchair ramp lowers the cost of each additional visit for an existing patient. For someone who has never established a physician relationship, the same ramp does nothing about the earlier problems of finding an affordable nearby provider, maintaining insurance coverage, or arranging time away from work. These are cost and search barriers, and the ADA's physical access mandate does not address them.

The implication for research design is direct. A study examining only the extensive margin, whether anyone saw a doctor at all would conclude the ADA had no effect on healthcare. A study examining only the intensive margin, how often people already in the system visited, would conclude it had a meaningful positive one. Both conclusions are partially true. Decomposing into margins is not a methodological nicety here; it is the only way to read what the law actually did.

Robustness Checks

Table 6 reports results across ten alternative specifications for each outcome. The extensive margin null is stable throughout. Age-subgroup regressions, which re-estimate the model separately for three age bands to check whether the null is masking effects concentrated in one part of the life cycle, confirm it holds broadly: the preferred estimate is −0.0012 (SE = 0.0060) among 25–55 year-olds, 0.0065 (SE = 0.0203) among 18–24 year-olds, and −0.0037 (SE = 0.0082) among 56–64 year-olds; none approaches significance. Missing data sensitivity checks, which test whether the baseline result changes when the roughly 11 percent of observations missing `above_poverty` (a binary indicator for whether the respondent's income falls above the federal poverty line) are handled differently confirm the baseline is not sensitive to this choice: mean/mode imputation yields $\hat{\beta} = -0.0019$ ($p = 0.672$) and the missing-indicator method, which adds a separate dummy variable flagging each observation with a missing covariate rather than dropping it, yields $\hat{\beta} = -0.0005$ ($p = 0.907$). Both estimates fall within a narrow band around the baseline. Shifting the treatment cutoff to Post $\geq$ 1991 or Post $\geq$ 1992 to account for the possibility that provider compliance lagged behind the law's formal enactment changes neither the sign nor the significance of the extensive margin estimate. A placebo test, which re-estimates the DiD on pre-period data only (1983–1989) using a false treatment year of 1987 to check whether the design would spuriously detect an effect even where none should exist, yields $\hat{\beta} = -0.0011$ ($p = 0.87$), confirming that no latent pre-trend is generating the baseline null.

The positive intensive margin effect survives the same battery. Timing variants confirm it is not an artefact of the enactment date: Post $\geq$ 1991 yields $\hat{\beta} = +0.77$ ($p = 0.005$) and Post $\geq$ 1992 yields $\hat{\beta} = +0.65$ ($p = 0.022$), suggesting the effect persists regardless of when exactly compliance is assumed to begin. Age-subgroup analysis reveals the effect is concentrated among older working-age adults (56–64: $\hat{\beta} = +1.34$, $p = 0.002$) and young adults (18–24: $\hat{\beta} = +2.38$, $p = 0.022$), while the prime working-age group (25–55) shows no significant intensive margin effect a pattern consistent with those groups facing the largest physical access barriers in the first place. The intensive margin placebo test returns $\hat{\beta} = +0.3971$ ($p = 0.291$), ruling out a spurious pre-trend as the driver of the post-ADA positive estimates. The extended sample (1983–2000, using `LAMTWRK`, a separate NHIS variable measuring whether a respondent is limited in the kind or amount of work they can do due to a health condition, as a fallback disability classifier for post-1996 observations) yields $\hat{\beta} = +0.77$ ($p = 0.005$); this result is exploratory given that `LAMTWRK` and `LATOTAL` measure disability in conceptually different ways, and it is not interpreted causally.

Descriptive Evidence: Outcomes Without a Pre-ADA Baseline

Two outcomes lack a pre-ADA baseline and cannot support causal difference-in-differences estimation. They are reported as descriptive cross-sections in Table 7; no causal interpretation is claimed.

`has_regular_doc` (available 1994–1996 only): this variable equals 1 if the respondent identifies a doctor's office or clinic as their usual place of care, a rough proxy for having an established physician relationship. In the post-ADA period, severely disabled individuals were consistently less likely to have such a regular source of care than the moderately limited, with a gap of −7.5 pp in 1994 that narrows to −2.7 pp by 1996. The convergence is descriptively suggestive, but without a pre-1990 baseline there is no way to know whether it reflects the ADA, a secular trend, or something else entirely.

`delayed_care` (available 1993–1996 only): this variable equals 1 if the respondent reports having delayed seeking medical care due to cost in the past year, a direct measure of whether financial barriers are preventing healthcare access. The severely disabled reported a +4.1 pp higher rate of cost-driven delay than the moderately limited in 1993, a gap that essentially closes by 1994–1996, ranging from +0.7 to −0.7 pp. Again, no causal claim can be made. That cost-driven delay remains elevated in the earliest available post-ADA year is, however, consistent with the extensive margin null: if cost is the binding constraint on initial access, a physical accommodation mandate would not be expected to relieve it.

Conclusion

Using a difference-in-differences design and seven years of pre-ADA NHIS data, this paper finds that the ADA did not significantly increase the probability that severely disabled Americans saw a physician in a given year ($\beta = -0.0001$, $p = 0.99$). It did increase annual physician visits by approximately 0.77 an estimate that strengthens as controls are added and is significant at the 5 percent level suggesting the law deepened care for those already within the system. The parallel trends assumption holds comfortably at $F = 0.276$ ($p = 0.949$), lending credibility to the design. The pattern that emerges is one of margin asymmetry: the ADA appears to have made it easier to keep seeing a doctor, but not easier to start.

That asymmetry has a direct implication for disability health policy. The physical access mandates the ADA required ramps, accessible examination equipment, interpreter services reduced the cost of returning to a provider you already knew. They did not reduce the cost of finding and establishing care with one for the first time. If insurance coverage, provider search costs, or income constraints are the binding barriers to initial healthcare entry for severely disabled individuals, then physical access mandates are not the right instrument to close that gap. Closing it likely requires something on the insurance side: expanded Medicaid eligibility, outreach programmes that actively connect severely disabled individuals without a regular provider, or subsidised care coordination. These are not modest additions to the ADA's framework they are a different policy lever entirely and the results here suggest they deserve attention if the goal is to bring the severely disabled into the healthcare system rather than simply improve their experience once inside it.

Three limitations are worth stating plainly. First, the within-disability design compares severity groups both of which are covered by the ADA. If the moderately limited also gained from the law and there is no reason to assume they did not — the DiD understates the true effect for the severely disabled. Second, the NHIS is cross-sectional. Composition shifts within severity groups across years such as marginal workers being reclassified into `LATOTAL` = 1 by deteriorating economic conditions cannot be ruled out without panel data. Third, two of the three originally planned outcomes, whether an individual has a regular source of care and whether they delayed needed care exist only for the post-ADA years in this dataset and cannot be causally identified.

On external validity, three boundaries deserve honest acknowledgment. The analysis sample comes from the NHIS Activity Limitation Supplement, administered to roughly 10 percent of NHIS respondents. Supplement selection is not uniform: the selection rate is 17.8 percent among individuals aged 55–64 but only 3.7 percent among those aged 18–29, meaning the sample skews toward near-retirement adults. The results are more representative of older severely disabled individuals than of young disabled Americans, and applying the findings to younger age groups requires caution. The post-ADA window also covers only 1990 to 1996 the first six years of the law during which many providers were still working toward compliance. The intensive margin estimates in particular may understate the long-run effect, since full physical accessibility for existing buildings took considerably longer to achieve than the statute's initial deadlines contemplated. Finally, because the control group is moderately disabled rather than non-disabled, the estimates capture the ADA's effect on the severely disabled relative to the moderately limited. They are not directly comparable to studies contrasting disabled and non-disabled populations, and should not be read as the total effect of the law on disabled Americans as a whole. Within those boundaries older severely disabled adults in the early 1990s, benchmarked against those with moderate limitations the results are credible and internally valid.

Three extensions would materially sharpen what this paper can say. Linking disability severity records to MEPS expenditure data would reveal whether the 0.77 additional annual visits represent substantive specialist care or brief, low-cost encounters a distinction that matters considerably for assessing the law's health impact. State-level variation in ADA enforcement intensity, proxied by EEOC complaint rates or Department of Justice settlement records, could serve as a continuous treatment variable in a state-by-year panel design, directly testing whether the mechanism is compliance rather than contemporaneous policy change. And if insurance coverage is the binding constraint on first entry into care, linking disability severity to insurance status and utilization in MEPS data would test that mechanism directly and would tell policymakers whether the problem is access, cost, or both, and what instrument each calls for.

References

Acemoglu, D., & Angrist, J. D. (2001). Consequences of employment protection? The case of the Americans with Disabilities Act. *Journal of Political Economy*, *109*(5), 915–957. https://doi.org/10.1086/322836

Blewett, L. A., Rivera Drew, J. A., King, M. L., Williams, K. C. W., Backman, D., Chen, A., & Richards, S. (2024). *IPUMS Health Surveys: National Health Interview Survey, Version 7.4* [Data set]. IPUMS. https://doi.org/10.18128/D070.V7.4

Bound, J., & Waidmann, T. (2002). Accounting for recent declines in employment rates among the working-aged disabled. *Journal of Human Resources*, *37*(2), 231–250. https://doi.org/10.2307/3069608

Centers for Disease Control and Prevention. (2024). *Behavioral Risk Factor Surveillance System annual survey data*. U.S. Department of Health and Human Services. https://www.cdc.gov/brfss/annual_data/annual_data.htm

DeLeire, T. (2000). The wage and employment effects of the Americans with Disabilities Act. *Journal of Human Resources*, *35*(4), 693–715. https://ideas.repec.org/a/uwp/jhriss/v35y2000i4p693-715.html

Iezzoni, L. I., McCarthy, E. P., Davis, R. B., & Siebens, H. (2000). Mobility impairments and use of screening and preventive services. *American Journal of Public Health*, *90*(6), 955–961. https://doi.org/10.2105/ajph.90.6.955

Krahn, G. L., Hammond, L., & Turner, A. (2006). A cascade of disparities: Health and health care access for people with intellectual disabilities. *Mental Retardation and Developmental Disabilities Research Reviews*, *12*(1), 70–82. https://doi.org/10.1002/mrdd.20098

Krahn, G. L., Walker, D. K., & Correa-De-Araujo, R. (2015). Persons with disabilities as an unrecognized health disparity population. *American Journal of Public Health*, *105*(S2), S198–S206. https://doi.org/10.2105/AJPH.2014.302182

LaPlante, M. P. (1988). *Data on disability from the National Health Interview Survey, 1983–1985*. National Institute on Disability and Rehabilitation Research.

Ordway, A., Garbaccio, C., Richardson, M., Matrone, K., & Johnson, K. L. (2021). Health care access and the Americans with Disabilities Act: A mixed methods study. *Disability and Health Journal*, *14*, 100967. https://doi.org/10.1016/j.dhjo.2020.100967

Parker Harris, S., Gould, R., Bowers, A., Fujiura, G., & Jones, R. (2017). *Knowledge Translation Center systematic review: The ADA and health care—Final report*. ADA National Network. http://adata.org/national-ada-systematic-review

Parker Harris, S., Gould, R., & Caldwell, K. (2019). *The ADA and health care: Research brief*. ADA Knowledge Translation Center. https://adata.org/research_brief/ada-and-health-care

U.S. Department of Justice. (1992). *Title III technical assistance manual covering public accommodations and commercial facilities*. https://www.ada.gov/taman3.html

Sommers, B. D., Blendon, R. J., Orav, E. J., & Epstein, A. M. (2014). *Changes in utilization and health among low-income adults after Medicaid expansion or premium subsidies*. *JAMA Internal Medicine*, *174*(10), 1568–1574. https://pubmed.ncbi.nlm.nih.gov/27532694/

Iezzoni, L. I., Killeen, M. B., & O'Day, B. L. (2006). *Rural residents with disabilities confront substantial barriers to obtaining primary care*. *Health Services Research*, *41*(4), 1258–1275. https://pubmed.ncbi.nlm.nih.gov/16899006/

Peikes, D., Chen, A., Schore, J., & Brown, R. (2009). *Effects of care coordination on hospitalization, quality of care, and health care expenditures among Medicare beneficiaries*. *JAMA*, *301*(6),

Appendix

Table 1: Sample Construction

Panel A. Exclusion Waterfall

Step	Excluded	Remaining
Full NHIS extract, 1983–1996	—	1,130,227
Not administered Activity Limitation Supplement	−1,041,413	88,814
Analysis sample (LATOTAL = 1 or 2)	—	**88,814**
Treated: LATOTAL = 1 (severely disabled)	—	42,294
Control: LATOTAL = 2 (moderately limited)	—	46,520

Panel B. Outcome Coverage by Year (% non-missing)

Year	N	saw_doctor	dv12_clean	has_regular_doc	delayed_care
1983	6,365	100.0%	100.0%	**ABSENT**	**ABSENT**
1984	6,107	100.0%	100.0%	**ABSENT**	**ABSENT**
1985	5,349	99.7%	99.6%	**ABSENT**	**ABSENT**
1986	3,666	100.0%	100.0%	**ABSENT**	**ABSENT**
1987	6,908	99.8%	99.8%	**ABSENT**	**ABSENT**
1988	6,974	100.0%	100.0%	**ABSENT**	**ABSENT**
1989	6,879	99.3%	99.3%	**ABSENT**	**ABSENT**
1990	6,796	99.7%	99.7%	41.2%	**ABSENT**
1991	6,931	99.8%	99.7%	43.6%	**ABSENT**
1992	7,912	100.0%	99.9%	11.7%	**ABSENT**
1993	7,137	99.6%	99.6%	46.3%	51.8%
1994	7,400	99.2%	99.2%	80.6%	91.1%
1995	6,443	99.0%	99.0%	81.6%	91.0%
1996	3,947	98.8%	98.8%	86.0%	96.5%

Notes: The 1,041,413 excluded respondents were not administered the supplement in their survey year, a rotating survey design feature, not item nonresponse. Respondents coded LATOTAL = 0 (not in universe) are excluded because their disability status is unobserved. In Panel B, ABSENT indicates the variable was not collected in that survey year. TYPPLSICK (`has_regular_doc`) was absent from the NHIS supplement before 1990. DELAYCOST (`delayed_care`) was introduced in 1993, three years after the ADA. Neither variable has a pre-ADA baseline; both are reported as descriptive cross-sections only, with no causal interpretation. Shaded rows indicate the pre-ADA period.

Table 2: Variable Descriptions

Variable	Source	Definition	Role
saw_doctor	NHIS DV12 / VISITYRNO	= 1 if any physician visit in past 12 months	Outcome: extensive margin
dv12_clean	NHIS DV12, capped ≤ 96	Count of physician visits in past 12 months	Outcome: intensive margin
LATOTAL	NHIS Supplement	1 = unable to perform major activity; 2 = limited in kind/amount	Treatment assignment
post_1990	Derived	= 1 if survey year ≥ 1990	Post-treatment indicator
age	NHIS	Age in years (continuous)	Control
sex_male	NHIS	= 1 if male	Control
health_status	NHIS	Self-reported health (1–5 scale)	Control
region	NHIS	Census region (4 categories)	Control
marital_status	NHIS	Marital status (categorical)	Control
above_poverty	NHIS	= 1 if income above poverty line	Control — 12.5% missing
any_insurance	NHIS	= 1 if insured	Control — 46.0% missing
education	NHIS	Highest education level	97.0% missing — excluded
employed	NHIS	Employment status	99.5% missing — excluded
perweight	NHIS	Person-level sampling weight	Used in all WLS regressions

Notes: Blue rows = outcome variables. Yellow rows = treatment/post indicator. Red rows = variables excluded due to near-complete missingness in the Activity Limitation Supplement subsample.

Table 3: Summary Statistics: Pre-ADA Period (1983–1989)

Variable	Severely Disabled	Moderately Limited	Difference	p-value
saw_doctor (mean)	0.902	0.864	+0.038	0.000***
dv12_clean (mean visits/yr)	13.482	8.496	+4.986	0.000***
Age (mean years)	47.963	44.880	+3.082	0.000***
Share male	0.573	0.436	+0.138	0.000***
Share above poverty line	0.278	0.170	+0.108	0.000***
Share with insurance	Not available pre-1990	Not available pre-1990	—	—
Health status (mean, 1=exc., 5=poor)	3.873	3.110	+0.764	0.000***
N	**18,943**	**23,305**	—	—

Notes: Means are unweighted. p-values from two-sided t-tests assuming unequal variances. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Health status coded 1 = excellent, 5 = poor. Insurance variables (HIMCAID, HIWORK) were not consistently fielded in pre-1990 NHIS waves. All differences are statistically significant at the 1% level. The design relies on parallel trends, not level equivalence between groups.

Figure 1: **ADA Timeline**

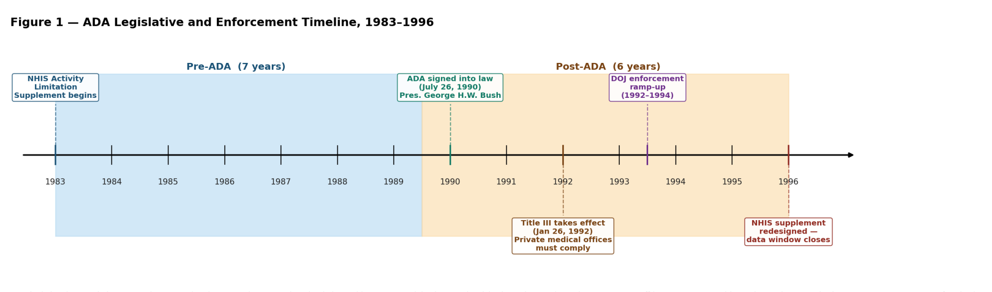
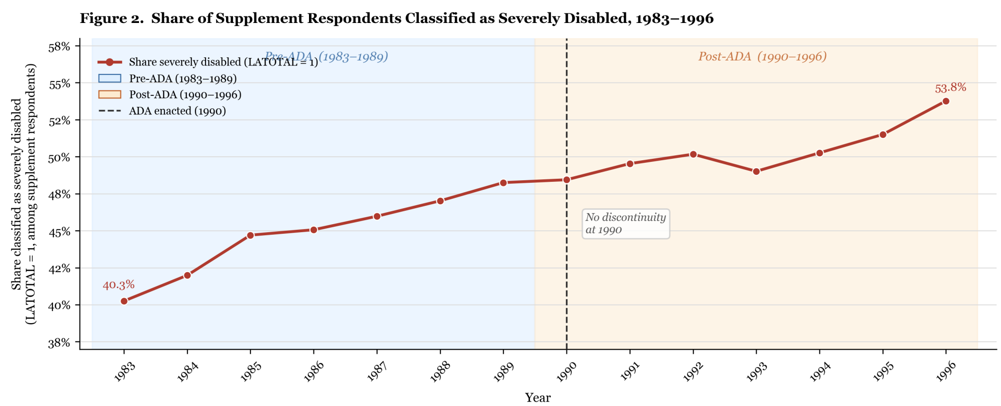
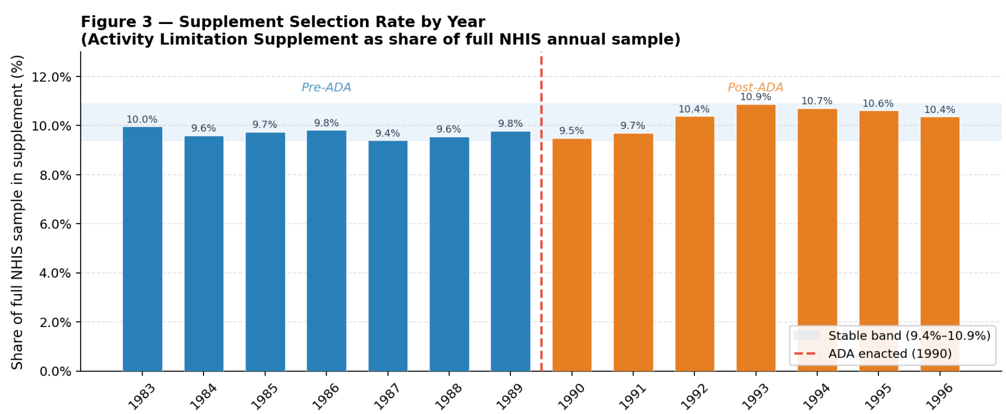


Figure 2: **Disability Severity Share**



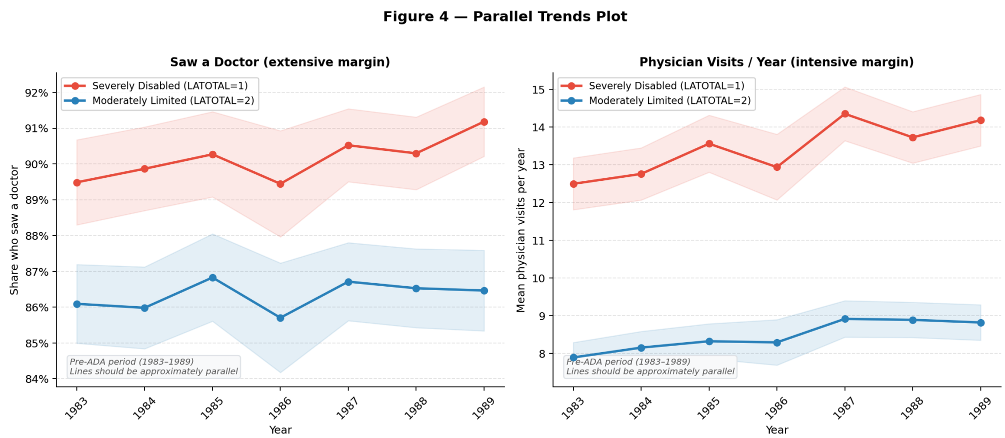
Note: The share of supplement respondents classified as severely disabled rises gradually from 40.3% (1983) to 53.8% (1996), with no visible discontinuity at 1990. A sharp jump at 1990 would suggest strategic reclassification — individuals shifting into the LATOTAL = 1 category after ADA enactment to obtain accommodation benefits. The smooth trend rules out this concern. Sample: NHIS Activity Limitation Supplement respondents with valid LATOTAL codes (1 or 2), 1983–1996.

Figure 3: **Supplement Selection**



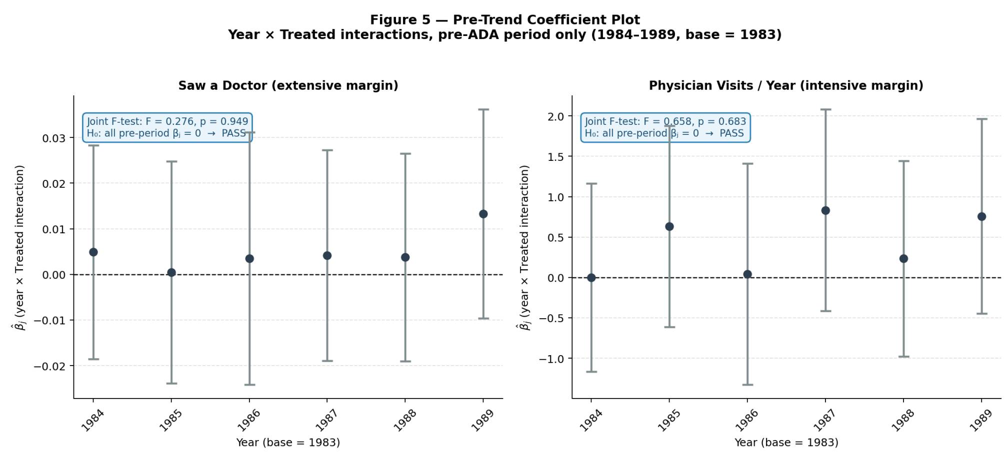
Notes: The supplement selection rate is stable at 9.4–10.9 percent across all study years, with no detectable shift at 1990. This rules out sampling composition changes as a source of the post-ADA DiD estimates.

Figure 4: **Parallel Trends**



Notes: Weighted group means (NHIS person weights) with 95% confidence intervals. Pre-ADA period 1983–1989 only. The approximately parallel trajectories of the two groups provide visual support for the identifying assumption before the formal F-test in Figure 5.

Figure 5: **Pre-Trend Coefficient Plot**



Notes: Each point is the coefficient on year × Treated from a WLS regression (NHIS person weights, HC3 robust SEs) estimated on the 1983–1989 pre-period only. Bars are 95% confidence intervals. All coefficients are close to zero and none is statistically distinguishable from zero (all p > 0.26). The joint F-test (p = 0.949) provides strong statistical support for the parallel trends identifying assumption.

Figure 6: **Event Study**

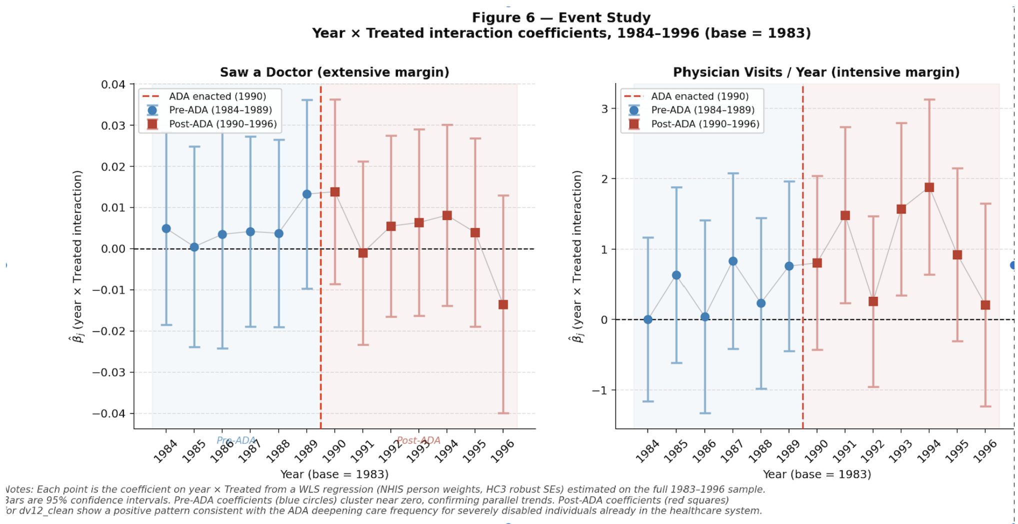


Figure 7: **2x2 DiD Decomposition**

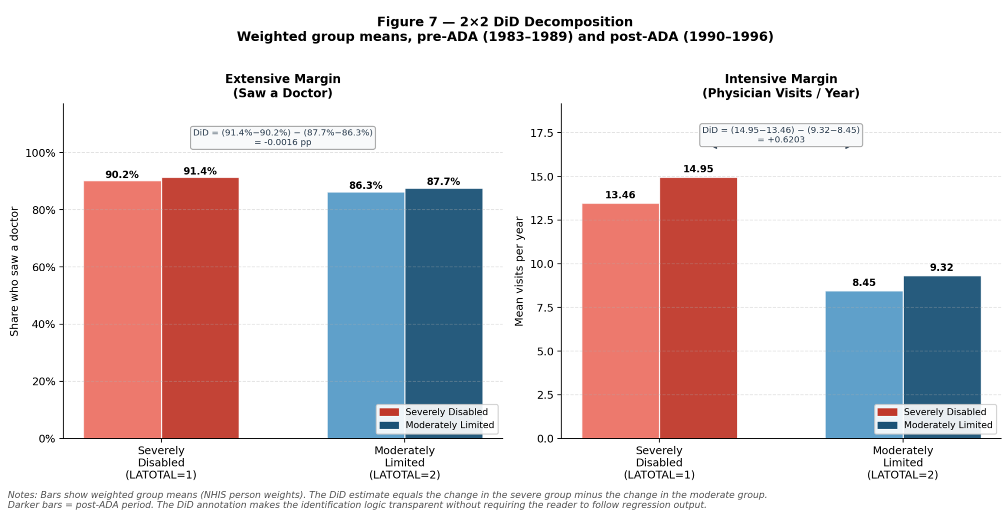


Table 4: DiD Results: Extensive Margin

Dependent variable: saw_doctor (= 1 if any physician visit in past 12 months)

	(1) Raw	(2) +Year FE	(3) +Controls
DiD coefficient	**-0.0016**	**-0.0012**	**-0.0001**
Std. Error	(0.0046)	(0.0046)	(0.0047)
p-value	0.734	0.795	0.988
N	88,505	88,505	78,008
R^2	0.0041	0.0042	0.0373
Year FE	No	Yes	Yes
Controls	No	No	Yes

Notes: WLS with NHIS person weights (PERWEIGHT). HC3 robust standard errors in parentheses. Treatment = LATOTAL=1 (unable to perform major activity); Control = LATOTAL=2 (limited in kind/amount); Post = year ≥ 1990. Model 3 controls: age, sex, self-reported health status, census region, marital status.

*** $p<0.01$ ** $p<0.05$ * $p<0.10$

Table 5: DiD Results- Intensive Margin
Dependent variable: physician visits per year

Panel A — dv12_clean (preferred estimate, capped at 96 visits)

	(1) Raw	(2) +Year FE	(3) +Controls
DiD coefficient	**+0.6203****	**+0.6641*****	**+0.7655*****
Std. Error	(0.2566)	(0.2564)	(0.2705)
p-value	0.016	0.010	0.005
N	88,485	88,485	77,990
R^2	0.0248	0.0253	0.0816
Year FE	No	Yes	Yes
Controls	No	No	Yes

Panel B — dv12 (uncapped, upper bound only)

	(1) Raw	(2) +Year FE	(3) +Controls
DiD coefficient	**+4.7488*****	**+4.7277*****	**+2.8746*****
Std. Error	(1.0173)	(1.0019)	(0.9707)
p-value	0.000	0.000	0.003
N	88,814	88,814	78,239
R^2	0.0076	0.0101	0.0191
Year FE	No	Yes	Yes
Controls	No	No	Yes

Notes: WLS with NHIS person weights (PERWEIGHT). HC3 robust standard errors in parentheses. Treatment = LATOTAL=1; Control = LATOTAL=2; Post = year ≥ 1990. Model 3 controls: age, sex, health status, region, marital status. Panel A preferred: values ≥97 are NHIS administrative codes, not genuine visit counts.

*** p<0.01 ** p<0.05 * p<0.10

Table 6: Robustness Checks Summary

Panel A — saw_doctor (extensive margin, binary)

Specification	DiD Coef.	Std. Error	p-value	N
Baseline Model 3	**-0.0001**	**(0.0047)**	**0.988**	**78,008**
Age 25–55	-0.0012	(0.0060)	0.843	49,420
Age 18–24	+0.0065	(0.0203)	0.747	5,550
Age 56–64	-0.0037	(0.0082)	0.651	23,038
Mean/mode imputation	-0.0019	(0.0045)	0.672	88,505
Missing-indicator	-0.0005	(0.0045)	0.907	88,505
Post $\geq$ 1991	-0.0043	(0.0047)	0.359	78,008
Post $\geq$ 1992	-0.0032	(0.0048)	0.512	78,008
Placebo year 1987	-0.0011	(0.0068)	0.875	37,314
Extended 1983–2000	-0.0303***	(0.0042)	0.000	244,018

Panel B — dv12_clean (intensive margin, preferred)

Specification	DiD Coef.	Std. Error	p-value	N
Baseline Model 3	**+0.7655*** **	**(0.2705)**	**0.005**	**77,990**
Age 25–55	+0.0450	(0.3677)	0.903	49,409
Age 18–24	+2.3759**	(1.0343)	0.022	5,548
Age 56–64	+1.3371***	(0.4244)	0.002	23,033
Mean/mode imputation	+0.6604***	(0.2499)	0.008	88,485
Missing-indicator	+0.7041***	(0.2497)	0.005	88,485
Post $\geq$ 1991	+0.7700***	(0.2750)	0.005	77,990
Post $\geq$ 1992	+0.6538**	(0.2859)	0.022	77,990
Placebo year 1987	+0.3971	(0.3761)	0.291	37,305
Extended 1983–2000	+0.7655***	(0.2705)	0.005	77,990

Notes: All specifications use WLS (NHIS person weights) and HC3 robust SEs. Baseline controls: age, sex, health status, region, marital status. Placebo row (red): DiD estimated on 1983–1989 pre-period only with false treatment year 1987 — should be insignificant if no pre-trend exists. Extended 1983–2000 uses LAMTWRK as fallback disability classifier — exploratory only.

*** p<0.01 ** p<0.05 * p<0.10

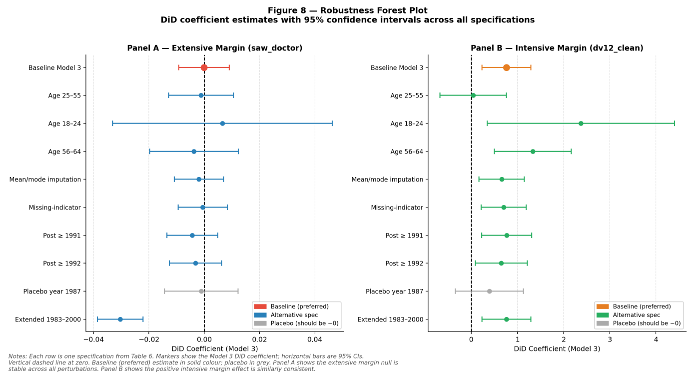


Figure 8: Robustness Forest Plot

Table 7: Descriptive Cross-Sections: Unidentified Outcomes

NO CAUSAL INTERPRETATION IS CLAIMED FOR THESE FIGURES

Panel A — has_regular_doc: usual source of care is a doctor's office or clinic
(available 1994–1996 only; no pre-ADA baseline exists)

Year	Severely Disabled (LATOTAL=1)	Moderately Limited (LATOTAL=2)	Gap (pp)
1994	81.5%	89.1%	**-7.5 pp**
1995	87.0%	91.2%	**-4.2 pp**
1996	87.4%	90.1%	**-2.7 pp**

Panel B — delayed_care: respondent delayed seeking care due to cost in past year
(available 1993–1996 only; introduced 3 years after ADA enactment)

Year	Severely Disabled (LATOTAL=1)	Moderately Limited (LATOTAL=2)	Gap (pp)
1993	78.0%	73.8%	**+4.1 pp**
1994	77.9%	77.2%	**+0.7 pp**
1995	79.2%	79.1%	**+0.2 pp**
1996	78.1%	78.8%	**-0.7 pp**

Notes: Unweighted group means for LATOTAL=1 (severely disabled) and LATOTAL=2 (moderately limited). Gap = Severely Disabled − Moderately Limited. Red = severe group has lower rate; green = severe group has higher rate. Neither variable has pre-ADA data; DiD estimation is not possible. These figures are descriptive only.